



ILLUSTRATION: Rami

Watch the boundaries blur between what is real and what is virtual as games get more immersive

Get into the game!

John was cornered. Chased down by the monster, he had taken desperate refuge in the wrecked boiler room, which was quickly filling up with steam. He had killed some of them, but this particular one had a very nasty habit of turning invisible.

He had two bullets left. Sweat stung his eyes and his wounds throbbed with pain, but he dared not blink. Then, out of the corner of his eyes, he thought he saw the steam move. Not sure. And again, no doubt this time—a shot from his hip was answered by an unearthly scream. John grimaced. He knew he had got the creature, but only a bullet to the head would kill it for good. One bullet left. He waited. The steam swirled, the creature was stirring. Wait. Slow, deliberate steps brought the swirling white ever closer to John. Wait. The thing picked up pace, and the haze wrapped itself around the invisible form, granting it a grotesque shape and rendering it visible. Wait. The very air seemed to pounce, vapour danced and twisted to reveal jaws from hell. Wait. The figure was now inches away from John, he could smell its foul breath... BANG! Slowly, hesitantly, he opened his eyes to a sight of relief—a lifeless abomination lay at John's feet. He had lived to tell the tale.

Thanks to the advances in technology, the adventures of John are no longer confined to some twisted plot in a book, or some cheesy scene in a movie, but are brought kicking and screaming to digital life and on your computer screen.

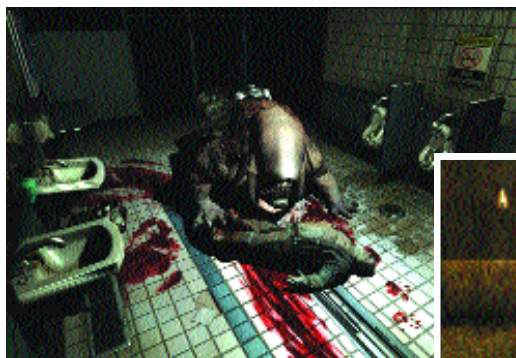
Prettier than a picture

Every industry under the sky is in search of its Holy Grail, something that will define the zenith of achievement for that sector. For the gaming business the goal has always been providing an experience so life-like, that the line between real and virtual is not just blurred, but is completely erased. Immersion and interactivity have been greatly blessed by the continual evolution of hardware. CPUs are faster and sound has moved from the era of the PC speaker to the realm of six-channel Dolby Digital surround audio. The greatest change though has been in the video card industry. Heralded by nVidia, video cards are becoming increasingly programmable and thanks to the collective efforts of both the hardware and the developer segments, graphics standards such as DirectX and OpenGL are moving towards becoming higher-level

languages, providing greater freedom and compatibility to game developers. These technological advances have brought the industry within striking distance of its goal.

Just how will this immersion be achieved? The first step is to pump up the basic building block of any game—the polygons. Greater the polygon count, the more detailed the world and everything in it. Polygon counts are no longer restricted to a few thousand, as the latest video cards are more than capable of handling a scene with 30 to 50 thousand polygons on the screen. Developers of *The Elder Scrolls III: Morrowind*, the third part in a series of role-playing games by Bethesda Softworks, have used this fact to create a highly detailed environment for the game. The on-screen characters are three-dimensional, created with high polygon counts and are clothed with multiple texture maps to make them look

much more believable and life-like. A character can thus be shown in meticulous armour, and the armour can either have a realistic metallic shine, or a rusty worn-out look.



Greater immersion in games has been successfully achieved by upping the polygon count and adding lighting effects, as seen here in *DOOM III* (above) and *Morrowind* (right)

